

# Cost-Asymmetric Physics-Informed Forecasting for Robust NHS Critical-Care Surge Allocation

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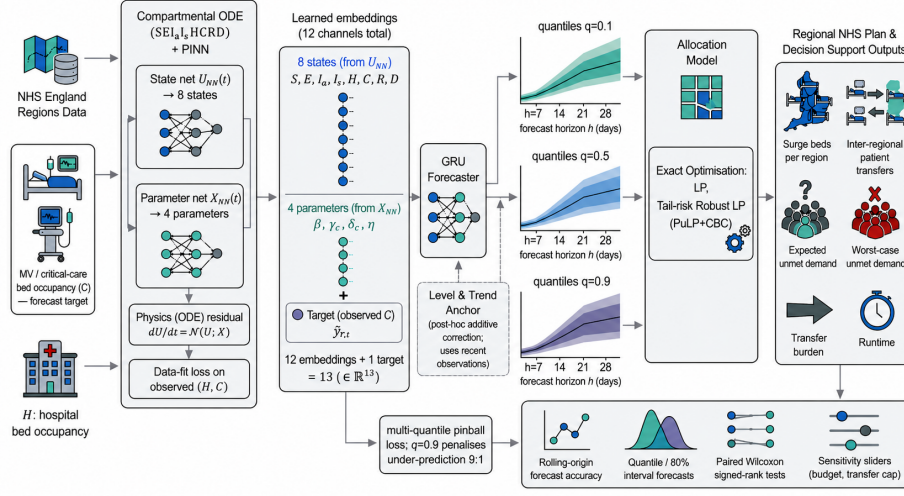
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**Abstract.** NHS critical-care planning during epidemic waves must pre-position surge capacity and route patients across regions before demand is known. We propose a forecast-to-decision pipeline that couples a per-region physics-informed forecaster on the  $SEI_a I_s HCRD$  system to a tail-risk robust linear program (LP) for bed-surge allocation. The forecaster is trained with a cost-asymmetric multi-quantile pinball loss whose  $q^{(0.9)}$  branch penalises under-prediction nine-to-one, together with a learnable level-and-trend anchor and a per-origin recalibration that draws on past errors alone to remove a long-horizon over-prediction bias. On a seven-region NHS England case study (Aug 2020 to Aug 2022) it cuts 14-day root-mean-square error (RMSE) by 25% over per-origin refitted autoregressive integrated moving average (ARIMA) and by 42% over a neural gated recurrent unit (GRU), and is competitive across all horizons (best at 7–21 days and level with refitted ARIMA at 28, where seasonal-naïve is strongest). Its quantile forecasts drive a hybrid-regional allocation solved exactly as an LP. With unconstrained transfers all budget-exhausting policies tie on coverage and the tail-risk robust formulation wins on transfer burden; once transfers are constrained the tie breaks and the proportional heuristics concede up to 60% more expected unmet demand. A sensitivity analysis over the forecaster, budget, tail-risk weight, travel cap, and transfer cap shows the budget to be the dominant control.

**Keywords:** physics-informed neural networks, cost-asymmetric forecasting, pinball loss, robust optimisation, linear programming, NHS England

## 1 Introduction

NHS critical-care surge planning during epidemic waves balances the cost of pre-positioning capacity against the risk of unmet demand while keeping inter-regional transfers manageable across the seven NHS England regions. COVID-19 placed sustained pressure on adult mechanical-ventilation capacity, making this a coupled forecasting and allocation problem under regime shift. Existing forecasting work spans statistical and recurrent baselines [2], graph-based epidemic and hospital-demand models [8,7], and physics-informed neural networks (PINNs) [14,1,13];



**Fig. 1.** Pipeline for NHS critical-care surge planning, from regional data and physics-informed forecasting to scenario-based robust allocation and evaluation.

interval and quantile evaluation is also standard in epidemic forecast hubs [4]. The closest intensive-care-unit (ICU) occupancy comparator, a bidirectional long short-term memory (LSTM) network with lagged mobility features and Monte-Carlo-dropout intervals on Lazio data [15], like most such work targets a single health system rather than multi-region planning. These forecasters are usually optimised for symmetric statistical accuracy, although critical-care planning has an asymmetric error cost: under-prediction risks stranded patients, whereas over-prediction wastes prepared capacity.

On the allocation side, robust facility-location and health-resource models handle demand uncertainty with mixed-integer linear programming (MILP), two-stage optimisation, and metaheuristics [6,16]. Bertsimas et al. [3] couple DELPHI dynamics to a bilinear mixed-integer program (MIP) for vaccination-site location, while Luo and Stellato [10] embed neural-ODE compartmental dynamics inside a McCormick-relaxed facility-location model. Our setting differs from Smart Predict-then-Optimize [5]. The forecaster is not trained through allocation regret, but its multi-quantile pinball objective is cost-asymmetric and feeds a robust optimisation layer. To our knowledge, no prior work combines physics-informed forecasting of NHS regional critical-care demand with cost-asymmetric quantile training and an exact robust linear-programming allocation across the seven NHS England regions.

This paper presents a forecast-to-decision pipeline (Figure 1) in four phases. *Phase A* pre-trains a per-region PINN on the  $SEI_a I_s HCRD$  ordinary-differential-equation (ODE) system; its twelve outputs (eight compartmental states, four transmission and clinical parameters) are frozen and passed to a temporal GRU head that emits the horizon-wise quantiles  $q^{(0.1)}$ ,  $q^{(0.5)}$ ,  $q^{(0.9)}$  under a multi-quantile pinball loss whose  $q^{(0.9)}$  branch penalises under-prediction nine-to-one, with a level-and-trend anchor. *Phase B* discretises these quantiles into three

scenarios with weights  $\pi = (0.2, 0.6, 0.2)$ . *Phase C* solves the allocation exactly as a linear program (deterministic and tail-risk-robust) via PuLP against three heuristic baselines on a shared LP-slave oracle. *Phase D* reports operational metrics and a sensitivity analysis over the forecaster, budget  $B$ , tail-risk weight  $\lambda_3$ , travel cap  $\tau$ , and transfer cap  $\kappa$ .

The main contributions are as follows: (1) **Cost-asymmetric forecaster.** A multi-quantile pinball loss, a level-and-trend anchor, and a per-origin recalibration reduce  $h=14$  RMSE by 25% relative to a refitted ARIMA model and 42% relative to a per-region GRU, and make the model competitive across all horizons (best at  $h=7-21$  and level with refitted ARIMA at  $h=28$ , where seasonal-naive leads). (2) **Robust LP allocation.** With unconstrained transfers the hybrid-regional robust LP ties every budget-exhausting policy on coverage and wins on transfer burden ( $4.6\times$  below population-proportional); constraining transfers (sparser network or capped transfer service) breaks the tie, the proportional rules concede up to 60% more expected unmet demand, and the robust LP is never beaten on coverage.

## 2 Forecasting Module

### 2.1 SEIRD Compartmental Model

For each region  $r \in \mathcal{R}$  with population  $N_r$  we model the local epidemic with an eight-compartment  $SEI_aI_sHCRD$  system whose states are the susceptible  $S_r$ , exposed  $E_r$ , asymptomatic and symptomatic infectious  $I_{a,r}$  and  $I_{s,r}$ , hospitalised  $H_r$ , critical-care  $C_r$ , recovered  $R_r$ , and deceased  $D_r$  populations of region  $r$ , each a count of people and together summing to  $N_r$ . Figure 2 shows the flow topology, in which the distinct probabilities  $\omega$  ( $I_s \rightarrow H$ ) and  $\phi$  ( $H \rightarrow C$ ) disambiguate the two clinical thresholds. The system evolves as

$$\begin{aligned}\dot{S}_r &= -\beta_r S_r (I_{s,r} + I_{a,r}) / N_r + \eta_r R_r, & \dot{E}_r &= \beta_r S_r (I_{s,r} + I_{a,r}) / N_r - \alpha E_r, \\ \dot{I}_{s,r} &= \alpha \rho E_r - d_s I_{s,r}, & \dot{I}_{a,r} &= \alpha (1 - \rho) E_r - d_a I_{a,r}, \\ \dot{H}_r &= d_s \omega I_{s,r} - (d_H + \mu) H_r, & \dot{C}_r &= \phi d_H H_r - (\gamma_{c,r} + \delta_{c,r}) C_r, \\ \dot{R}_r &= d_s (1 - \omega) I_{s,r} + d_a I_{a,r} + (1 - \phi) d_H H_r + \gamma_{c,r} C_r - \eta_r R_r, \\ \dot{D}_r &= \mu H_r + \delta_{c,r} C_r,\end{aligned}\tag{1}$$

subject to the conservation constraint  $S_r + E_r + I_{a,r} + I_{s,r} + H_r + C_r + R_r + D_r = N_r$ , where a dot denotes a time derivative ( $\dot{S}_r = dS_r/dt$ ). The eight fixed clinical constants (values from [1], Table 1) are the latent-to-infectious (incubation) rate  $\alpha$ , the symptomatic fraction  $\rho \in [0, 1]$ , the symptomatic and asymptomatic infectious-removal rates  $d_s$  and  $d_a$ , the hospital-outflow rate  $d_H$ , the symptomatic hospitalisation probability  $\omega \in [0, 1]$  ( $I_s \rightarrow H$ ), the in-hospital mortality rate  $\mu$  ( $H \rightarrow D$ ), and the critical-care escalation probability  $\phi \in [0, 1]$  ( $H \rightarrow C$ ); all rates have units of  $\text{day}^{-1}$ . The four time-varying, region-specific parameters learned from data are the transmission rate  $\beta_r$ , the critical-care recovery rate  $\gamma_{c,r}$  ( $C \rightarrow R$ ), the critical-care death rate  $\delta_{c,r}$  ( $C \rightarrow D$ ), and the immunity-waning rate  $\eta_r$  ( $R \rightarrow S$ ).

## 2.2 Per-Region PINN-SEIRD

The PINN consists of two multilayer perceptrons (MLPs) following [1]: a state network  $U_{NN}^r$  (five hidden layers of 20 units) and a parameter network  $X_{NN}^r$  for  $(\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r)$  (three hidden layers of 20 units), with tanh hidden activations and sigmoid outputs. The loss combines a data fit on the two observable compartments with an ODE residual against Equation (1). The observed compartments are  $H$  (daily general-and-acute hospital bed occupancy) and  $C$  (mechanical-ventilation [MV], i.e. critical-care, bed occupancy), the only states with daily regional NHS counts;  $C$  is the quantity the temporal head forecasts and the allocation consumes as demand  $\xi$ :

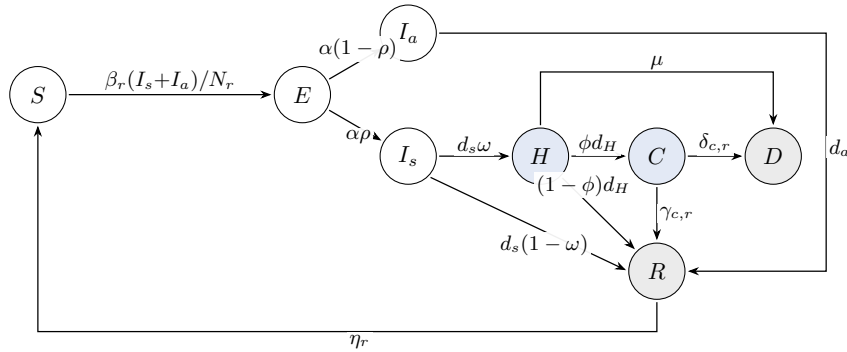
$$\begin{aligned} \mathcal{L}_r^{PINN} = & \sum_{t \in \mathcal{T}_{\text{train}}} \sum_{k \in \{H, C\}} (U_{NN,k}^r(t) - y_{r,t,k}/N_r)^2 \\ & + \lambda_{ode} \sum_{t \in \mathcal{T}_{\text{coll}}} \left\| \frac{dU_{NN}^r(t)}{dt} - \mathcal{N}(U_{NN}^r; X_{NN}^r) \right\|^2, \end{aligned} \quad (2)$$

where  $\mathcal{N}$  is the right-hand side of Equation (1), derivatives use PyTorch autograd at 256 collocation points sampled uniformly in  $\mathcal{T}_{\text{coll}}$ , and observations are population-normalised. We pre-train each region’s PINN for 1500 Adam steps with  $\lambda_{ode} = 0.1$ ,  $\text{lr} = 5 \times 10^{-3}$ ; both terms converge to the  $2\text{--}8 \times 10^{-6}$  range across regions.

With only  $H$  and  $C$  observed, the four learnable parameters  $(\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r)$  are not jointly identifiable ( $\gamma_{c,r}$  and  $\delta_{c,r}$  appear additively in  $\dot{C}_r$ ), so we make no per-trajectory interpretability claim: the four-channel output is consumed as a learned feature vector by the GRU, whose signal the “w/o PINN params” ablation confirms (Section 5.1;  $h=14$  RMSE  $14.46 \rightarrow 27.25$ ). Structural identifiability under  $\{H, C\}$  observation is future work.

## 2.3 Temporal Head with Level-and-Trend Anchor

Once pre-trained the PINN is *frozen*: at every origin, including those in the Omicron test period, its networks are *evaluated* (never refit) to supply features,



**Fig. 2.**  $SEI_a I_s H C R D$  compartmental model used by the physics-informed forecaster.

so any temporal extrapolation of the state network beyond its Alpha/Delta fit is absorbed and corrected downstream by the GRU and the anchor of Equation (3). For each lookback window of length  $L = 28$  ending at origin  $t$ , the augmented feature sequence is  $\mathbf{f}_{r,i} = [U_{NN}^r(i) \parallel X_{NN}^r(i) \parallel \tilde{y}_{r,i}] \in \mathbb{R}^{13}$ ,  $i \in \{t - L + 1, \dots, t\}$ . Here  $U_{NN}^r(i)$  contributes all eight compartments (including the never-observed latent  $S, E, I_a, I_s, R, D$ ), so the head receives the full physics-inferred state, and  $\tilde{y}_{r,i}$  is the target z-scored with training-period statistics only. A two-layer GRU (hidden 128, dropout 0.15) decodes the final hidden state via a per-horizon two-layer MLP (width 64, Gaussian-error-linear-unit [GELU] activations) into standardised quantile deltas  $\Delta_{r,t}^{h,q}$  for  $h \in \mathcal{H} = \{7, 14, 21, 28\}$  and  $q \in \mathcal{Q} = \{0.1, 0.5, 0.9\}$ .

A *level-and-trend anchor* makes the GRU’s job residual:

$$\hat{\zeta}_{r,t}^{h,q} = \Delta_{r,t}^{h,q} + a^{h,q} \tilde{y}_{r,t} + g^{h,q} v_{r,t}^{(7)} \left(\frac{h}{7}\right)^{\psi^q}, \quad (3)$$

where  $\tilde{y}_{r,t}$  is the most recent z-scored observation,  $v_{r,t}^{(7)} = \tilde{y}_{r,t} - \tilde{y}_{r,t-7}$  is the one-week slope, and  $a^{h,q}, g^{h,q}, \psi^q \in (0, 1)$  are learnable scalars. The Holt-Winters-style damping exponent  $\psi^q$  prevents slope-extrapolation overshoot at long horizons; we initialise  $\psi^{0.5} \approx 0.88$  (near-linear) and  $\psi^{0.1} = \psi^{0.9} = 0.5$  (square-root damping). Forecasts in the original scale follow by inverting standardisation.

## 2.4 Pinball Loss for Cost-Asymmetric Quantiles

The forecaster minimises the multi-quantile pinball loss [9]:

$$\mathcal{L}_{\text{pinball}} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbb{E}_{(t,h)} [\max(qe, (q-1)e)], \quad e = \tilde{y}_{r,t+h-1} - \hat{\zeta}_{r,t}^{h,q}, \quad (4)$$

evaluated at all  $(h, q) \in \mathcal{H} \times \mathcal{Q}$ . The  $q = 0.5$  pinball collapses to L1 (point accuracy); the  $q = 0.9$  pinball penalises under-prediction nine-to-one. Splitting the kernel into its two arms,  $\max(qe, (q-1)e) = q \max(e, 0) + (1-q) \max(-e, 0)$ , exposes it as a convex newsvendor surrogate of the allocation’s positive-part shortage cost, with the  $q=0.9$  asymmetry encoding that an under-bed risks a stranded patient whereas an over-bed only wastes pre-positioned capacity [5]. The forecaster is not trained through the allocation solver. We report  $\hat{y}^{0.5}$  as the point forecast and  $[\hat{y}^{0.1}, \hat{y}^{0.9}]$  as an 80% interval.

We train for up to 800 AdamW epochs (lr =  $10^{-3}$ , weight decay  $10^{-4}$ , cosine annealing) on one-day-stride Alpha-wave lookback windows (270 per region, 1,890 in total), with early stopping (patience 80) on the 1,134 Delta-wave validation windows’ pinball loss. Once selected, the model is frozen and applied to the Omicron test period via  $\Delta t = 7$ -day rolling origins, yielding 32 origin dates.

## 2.5 Per-Origin Recalibration

Applied frozen across a regime shift (high-demand Alpha/Delta training, mild Omicron test), the point forecast inherits a systematic over-prediction that

grows with horizon. We correct it with two steps. The first recalibrates the bias, subtracting at each rolling origin  $t$  and horizon  $h$  the mean of the model’s last  $K=3$   $h$ -step errors whose target date is already observed ( $< t$ ), the adaptation that a per-origin refit gives for free. The second blends with persistence, mixing the recalibrated forecast with seasonal-naive(7) under a per-horizon weight  $\alpha \in \{0, 0.1, \dots, 1\}$  fitted on an expanding window of realised origins (pure model,  $\alpha=1$ , until seven origin–region pairs are realised), so long horizons lean on persistence. Both use only information available at  $t$ ; the 80% interval is left as the quantile heads produce it, so the downstream allocation is unchanged.

### 3 Regional Allocation Model

The allocation module decides how many surge beds to place in each NHS region and how to route excess demand through inter-regional patient transfers, given the quantile forecasts of Section 2. We present a deterministic linear program, a tail-risk robust extension, and closed-form heuristic baselines.

#### 3.1 Sets, Parameters, and Decision Variables

The scope is *hybrid regional*: demand nodes and capacity sites coincide with the seven NHS England commissioning regions, one set  $\mathcal{R} = \{r_1, \dots, r_7\}$ . Unlike facility-location models that choose which sites to open [3], every region already operates critical care, so there is no opening decision and surge cost is uniform per bed; the model decides only bed counts and routing. Indices: regions  $r, r' \in \mathcal{R}$ ; horizons  $h \in \mathcal{H} = \{7, 14, 21, 28\}$  days; scenarios  $s \in \mathcal{S} = \{\text{low}, \text{median}, \text{high}\}$  from  $q^{(0.1)}, q^{(0.5)}, q^{(0.9)}$  with weights  $\pi = (0.2, 0.6, 0.2)$ , a representative-quantile discretisation of lower, central, and upper demand stress (a scenario mixture, not calibrated probability bins). Parameters: forecast demand  $\xi_{r,h}^s = \hat{q}_r^{(s)}(t_0 + h)$ ; baseline critical-care capacity  $\Gamma_r = 1.05 \times \text{Delta-peak}$  (the bed ceiling of region  $r$ ); per-region surge cap  $K_r = 0.5 \Gamma_r$ ; budget  $B = 0.20 \sum_r \Gamma_r$ ; great-circle inter-centroid distance  $\ell_{rr'}$  in km ( $\times 1.3$  road-detour); travel time  $T_{rr'} = 60 \ell_{rr'} / 80$  min; mutual-aid cap  $\tau = 240$  min; outbound transfer cap  $\kappa$  (displaced patients a region’s transfer service can sustain at a checkpoint; uncapped at the headline operating point). Decision variables fall in two layers. The *first-stage* (here-and-now) capacity decision, the planner’s only true control, is the surge allocation  $b_r \in [0, K_r]$ , shared across scenarios and horizons (beds stood up once serve every checkpoint). The per-scenario *second-stage* recourse is the flow  $z_{rr',h}^s \geq 0$  (defined for  $T_{rr'} \leq \tau$ ), whose self-loop  $z_{rr,h}^s$  is demand served *locally* at zero distance and whose off-diagonal ( $r' \neq r$ ) entries are inter-regional transfers, together with the unmet demand  $u_{r,h}^s \geq 0$ ; a tail auxiliary  $W \geq 0$  completes the robust variant. Of the 42 ordered region pairs, 24 satisfy  $T_{rr'} \leq \tau$  (12 undirected mutual-aid links); Midlands reaches every region, South West only Midlands.

### 3.2 Deterministic Median-Scenario LP

The deterministic model optimises the median scenario  $m$  ( $q^{(0.5)}$ ):

$$\min \sum_{r,h} u_{r,h}^m + \theta \sum_{r \neq r',h} \ell_{rr'} z_{rr',h}^m, \quad (5)$$

with  $\theta = 10^{-3}$  a transfer tie-break weight (one bed of weekly snapshot unmet outweighs 1000 patient-km of transfer). The objective minimises *unmet demand*; the second term only breaks ties between equal-coverage plans by preferring less patient transport, so the surge beds  $b$  are placed to reduce shortage, not to minimise transfers. Each  $u_{r,h}^m$  is a snapshot of bed-shortfall at the weekly checkpoint  $h$ , so the objective is the snapshot unmet summed over the four-week planning horizon, not a continuous bed-day integral. Constraints:

$$z_{rr,h}^m + \sum_{r' \neq r: T_{rr'} \leq \tau} z_{rr',h}^m + u_{r,h}^m \geq \xi_{r,h}^m \quad \forall r, h, \quad (6)$$

$$z_{rr,h}^m + \sum_{r' \neq r: T_{r'r} \leq \tau} z_{r'r,h}^m \leq \Gamma_r + b_r \quad \forall r, h, \quad (7)$$

$$\sum_{r' \neq r} z_{rr',h}^m \leq \kappa \quad \forall r, h, \quad (8)$$

$$\sum_r b_r \leq B, \quad (9)$$

$$z_{rr',h}^m = 0 \quad \text{if } T_{rr'} > \tau, \quad (10)$$

$$b_r, z_{rr',h}^m, u_{r,h}^m \geq 0. \quad (11)$$

Equation (6) is a demand balance: each region's forecast demand is served locally ( $z_{rr,h}^m$ ), transferred to a reachable region ( $z_{rr',h}^m$ ,  $r' \neq r$ ), or left unmet ( $u_{r,h}^m$ ). Equation (7) caps the beds a region uses (local plus transferred-in patients) at baseline plus surge; Equation (8) bounds outbound patients by the transfer-service capacity  $\kappa$ ; Equation (9) charges the budget once per region; Equation (10) suppresses transfers beyond the mutual-aid window. As  $u_{r,h}^m$  is minimised, Equation (6) holds it at the residual shortfall  $\max(0, \xi_{r,h}^m - \sum_{r'} z_{rr',h}^m)$ , a derived quantity rather than a free decision.

### 3.3 Tail-Risk Robust Objective

To hedge against the high-demand scenario the robust model imposes Equations (6) to (8) for every  $s \in \mathcal{S}$ , retains Equations (9) and (10), and adds the tail-risk term  $\lambda_3 W$  where  $W \geq \sum_{r,h} u_{r,h}^s$  for  $s \in \mathcal{S}_{\text{tail}} = \{\text{high}\}$ :

$$\min \sum_{s \in \mathcal{S}} \pi_s \left( \sum_{r,h} u_{r,h}^s + \theta \sum_{r \neq r',h} \ell_{rr'} z_{rr',h}^s \right) + \lambda_3 W. \quad (12)$$

$\lambda_3=0$  recovers the three-scenario risk-neutral objective;  $\lambda_3 \rightarrow \infty$  yields a worst-case policy on the high scenario. We report  $\lambda_3=1$  throughout. With a single tail scenario, minimisation drives  $W = \sum_{r,h} u_{r,h}^{\text{high}}$ , so  $\lambda_3 W$  is a linear penalty on the worst-scenario shortfall (equivalently, extra weight on the high scenario) rather than a Rockafellar–Uryasev conditional value-at-risk (CVaR); the auxiliary form generalises unchanged to multiple tail scenarios. Because the high scenario imposes  $q^{(0.9)}$  jointly across regions and horizons, the construction is co-monotone and its tail over-conservative relative to the true joint distribution; we treat this as an explicit stress-test choice and examine its efficiency cost in Section 5.3.



### 3.4 Exact LP Solution and Baselines

Both models are linear programs (7 surge variables, 84 self-loops, up to 504 transfer flows, 84 unmet variables, and the tail auxiliary, all continuous); a deployed plan rounds the per-region surge to integers, changing total coverage by at most a few beds at this budget. We solve them exactly in PuLP with the bundled COIN-OR Branch-and-Cut (CBC) solver; at seven-region scale each instance reaches global optimality in  $<0.1$  s, so metaheuristic search is deferred to the trust-level extension (Section 6).

Every policy is scored on equal footing by a common LP re-evaluation: for a fixed surge  $\mathbf{b}^*$  ( $\sum_r b_r^* \leq B$ ,  $b_r^* \leq K_r$ ), the residual problem in  $(z, u)$  fills in transfers and unmet demand across all three scenarios, giving honest  $E[u]$  and  $u^{\text{worst}}$  for the deterministic, robust, and heuristic allocations alike.

Finally, three closed-form heuristics give a no-optimisation floor: population-proportional uses ONS region populations, demand-proportional uses expected peak demand, and greedy shortage-first adds single-bed increments to the most-unmet region. Each fixes  $\mathbf{b}^*$ , completed by the same LP step.

## 4 NHS England Case Study

The case study uses two public sources covering 1 Aug 2020 – 31 Aug 2022 (the period of daily NHS regional publication): NHS England COVID-19 Hospital Activity [11], which provides daily hospital and mechanical-ventilation bed occupancy at NHS regional resolution; and ONS Mid-Year Population Estimates [12] on 2021 geography, used as the per-region PINN normaliser and the population-proportional baseline denominator. All experiments use the seven NHS England commissioning regions. The series is split chronologically by epidemic wave so the most clinically distinct wave is held out: training on 1 Aug 2020 – 31 May 2021 (Alpha and early vaccination), validation on 1 Jun – 30 Nov 2021 (Delta) for hyper-parameter selection and early stopping, and testing on 1 Dec 2021 – 31 Aug 2022 (Omicron and post-Omicron). Omicron’s case-to-hospitalisation and hospitalisation-to-critical-care ratios differ markedly from earlier waves, providing a regime-shift stress test.

For the allocation problem, baseline capacity  $\Gamma_r$  is *measured* as each region’s Delta-wave peak MV-bed occupancy  $\times 1.05$ , whereas the per-region surge cap  $K_r = 0.5 \Gamma_r$ , total budget  $B = 0.20 \sum_r \Gamma_r \approx 213$  beds (the surge headroom above the  $\sum_r \Gamma_r \approx 1065$ -bed measured baseline, not existing beds), mutual-aid travel cap  $\tau = 240$  min, and uncapped outbound transfers ( $\kappa = \infty$ ) are planning assumptions. The surge  $b_r$  is the model’s decision, the extra MV beds a region stands up above its measured baseline. The forecaster, budget  $B$ , tail weight  $\lambda_3$ , travel cap  $\tau$ , and transfer cap  $\kappa$  are each swept in Section 5.3. Forecast metrics are macro-averaged across the seven regions and the 32 rolling-origin evaluations spanning the Omicron test period; the full solver comparison is reported at the first Omicron decision origin (29 Dec 2021), with an all-origin check in Section 5.3. Code, configurations, and the cleaned dataset are available in the project repository.



**Table 1.** Forecast accuracy (best **bold**, second best underlined per column).

| Model                       | RMSE by horizon (days) |              |              |              | MASE        |
|-----------------------------|------------------------|--------------|--------------|--------------|-------------|
|                             | 7                      | 14           | 21           | 28           | (all $h$ )  |
| Seasonal-naive(7)           | 13.53                  | 18.16        | <u>22.31</u> | <b>25.98</b> | 0.63        |
| ARIMA per region            | 11.33                  | 17.80        | 23.41        | <u>28.02</u> | <u>0.61</u> |
| XGBoost (lag features)      | 12.45                  | 22.90        | 36.42        | 58.88        | 0.89        |
| GRU per region              | 14.13                  | 22.92        | 33.35        | 43.97        | 0.90        |
| PinnGRU (raw)               | <u>11.02</u>           | <u>14.46</u> | 24.19        | 34.63        | 0.77        |
| <b>PinnGRU+recal (ours)</b> | <b>10.56</b>           | <b>13.34</b> | <b>20.79</b> | 28.54        | <b>0.56</b> |

## 5 Results and Discussion

### 5.1 Forecasting Accuracy

Table 1 reports per-horizon RMSE and mean absolute scaled error (MASE) against four baselines under a common 32-origin rolling-origin protocol; MASE scales the mean absolute error (MAE) by the in-sample seasonal-naive(7) MAE, so values below 1 beat that benchmark. The *raw* PinnGRU point forecast (frozen PINN encoder of Section 2.2 with the GRU head of Section 2.3) beats all four baselines at  $h=7$  and  $h=14$ ; a paired Wilcoxon signed-rank test across the 32 origins confirms the  $h=14$  gain over ARIMA ( $-3.88$  MV beds,  $p=0.003$ ). It degrades at long horizons, however, where a regime-shift over-prediction bias (an  $h=28$  underestimation rate of only 10.3%) leaves it behind ARIMA and seasonal-naive ( $+5.53$  MV beds vs ARIMA at  $h=28$ ,  $p=0.006$ ). The per-origin recalibration of Section 2.5 removes this bias and blends with persistence at long horizons. The recalibrated model is best at  $h=7$ , 14 and 21, cutting the operational 14-day RMSE by 25% over ARIMA and 42% over the per-region GRU, and at  $h=28$  ties ARIMA (28.5 vs 28.0) with only seasonal-naive (26.0) still ahead. Its  $h=28$  underestimation rate rises to 37%, so the point forecast is now a well-calibrated median rather than a systematic over-predictor.

XGBoost and the per-region GRU, lag-only learners bounded by the 28-day window, cannot extrapolate into Omicron’s altered case-to-hospitalisation regime and degrade sharply ( $h=28$  RMSE 58.9, 44.0), whereas ARIMA’s per-origin refit degrades only modestly.

Three ablations against the raw PinnGRU ( $h=14$  RMSE 14.46, MASE 0.77) isolate the load-bearing parts: replacing the pinball loss with a mean-squared-error (MSE) loss lifts  $h=14$  RMSE to 18.70 (MASE 0.97), skipping PINN pre-training to 17.27 (MASE 0.76), and zeroing the four PINN parameter features to 27.25 (MASE 1.23), the largest regression. The physics prior, not capacity, drives the gain over the per-region GRU.

### 5.2 Allocation Performance Across Solvers

Table 2 compares the allocation policies on the seven-region problem with  $B = 213$  beds (20% of total baseline), four-hour mutual-aid cap, and PinnGRU

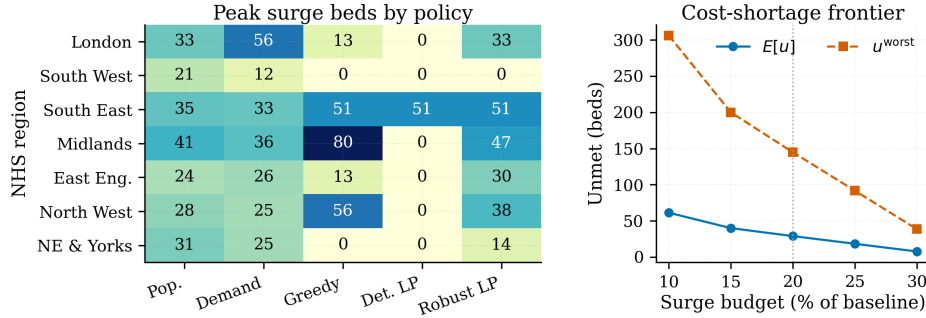
**Table 2.** Allocation comparison under  $B = 213$  beds,  $\tau = 240$  min, and three PinnGRU scenarios; the  $\kappa=10$  column re-scores each policy under a capped transfer service.

| Policy                                      | Beds  | $\kappa=\infty$ |                    | $\kappa=10$ | Transfer<br>(patient km) | Time<br>(s) |
|---------------------------------------------|-------|-----------------|--------------------|-------------|--------------------------|-------------|
|                                             |       | $E[u]$          | $u^{\text{worst}}$ | $E[u]$      |                          |             |
| Population-proportional                     | 212.7 | 29.0            | 144.8              | 48.8        | 3119.8                   | 0.00        |
| Demand-proportional                         | 212.7 | 29.0            | 144.8              | 51.4        | 2615.3                   | 0.00        |
| Greedy shortage-first                       | 212.7 | 29.0            | 144.8              | <b>32.3</b> | 1140.0                   | 9.98        |
| Deterministic LP                            | 51.4  | 83.3            | 416.4              | 84.4        | 1175.1                   | 0.03        |
| <b>Robust LP (<math>\lambda_3=1</math>)</b> | 212.7 | 29.0            | 144.8              | <b>32.3</b> | <b>671.7</b>             | 0.04        |

$q^{(0.1)}/q^{(0.5)}/q^{(0.9)}$  forecasts at the first Omicron-period origin; every allocation is re-scored by the common LP re-evaluation over the full three-scenario set, so  $E[u]$  and  $u^{\text{worst}}$  are on equal footing.

The *robust LP* achieves the lowest transfer burden (672 patient-km,  $4.6\times$  below population-proportional and  $1.7\times$  below greedy), but at  $\kappa=\infty$  this is a tie-break among policies with identical rounded  $E[u]$  and  $u^{\text{worst}}$ , not a coverage gain. The *deterministic LP* spends only 51 of the 213 beds and pays  $\sim 3\times$  higher  $E[u]$  and  $u^{\text{worst}}$ , quantifying the value of  $\pi_s$ -weighted expectation over median-only optimisation. The  $\kappa=10$  column previews Section 5.3, where a capped transfer service breaks the tie.

The per-region allocations underlying Table 2 (Figure 3a) show the deterministic LP surging only Midlands and South East (where median demand exceeds baseline), whereas budget-exhausting policies cover those two and differ on the rest, the robust LP compromising with non-trivial North West and London surge.

**Fig. 3.** Allocation structure and robust-LP budget frontier: (a) per-region surge beds by policy at the first Omicron origin; (b) expected and worst-case unmet demand as the surge budget grows.

### 5.3 Forecast-Quality and Parameter Sensitivity

Substituting each forecaster into the robust LP at the 29 Dec 2021 origin, the  $q^{(0.9)}$  peak sums (PinnGRU 1421, XGBoost 1105, GRU 973, ARIMA 875, seasonal-naive 745 beds) all exceed the realised peak (797) except seasonal-naive. Realised unmet is zero throughout, so forecast quality buys *efficiency* at the  $q^{(0.9)}$  margin rather than coverage in this mild Omicron instance.

Across all 32 Omicron origins the pattern holds. Robust LP, population-proportional, and demand-proportional allocations share a mean expected unmet of 2.9, but the robust LP cuts mean transfer burden to 76 patient-km versus 406 and 337. The proportional baselines fill the full 212.7-bed budget at every origin, whereas the robust LP scales surge to local demand (mean 35.4 beds), so the routing-efficiency gain holds across the full demand distribution.

Turning to the planning parameters, a one-at-a-time sweep isolates each control. The budget  $B$  dominates (Figure 3b). Cutting  $B/\sum_r \Gamma_r$  from 0.20 to 0.10 doubles  $E[u]$  and triples  $u^{\text{worst}}$ , while 0.30 cuts  $E[u]$  to  $\sim 1/4$ .  $\lambda_3$  is *non-binding* at  $B = 0.20$  (the expectation-minimising solution already clears the high tail); only a tighter budget ( $\leq 15\%$ ) re-engages it.

The transfer controls  $\tau$  and  $\kappa$  determine when placement matters. With uncapped recourse on the 12-link network the Table 2 coverage tie persists at 10–20% budgets: coverage depends only on *total* surge, so placement is fungible and optimisation buys routing alone. Constraining recourse breaks the tie. Pruning the network to five links ( $\tau=120$  min) puts the robust LP ahead of every heuristic on expected and worst-case unmet (29.9 vs 32.3–34.9 at  $B=20\%$ ), and capping the transfer service at  $\kappa=10$  gives the same separation (the  $\kappa=10$  column of Table 2), where greedy matches the LP only by issuing 213 oracle calls at  $1.4\times$  its transfer burden. Placement earns its keep when transfers are scarce—the operating regime of a real surge.

## 6 Conclusion

We presented a forecast-to-decision pipeline for NHS critical-care surge planning that couples a per-region physics-informed forecaster to a tail-risk robust allocation linear program. With a per-origin recalibration, PinnGRU is competitive across the horizons that drive surge decisions, cutting 14-day RMSE by 25% over refitted ARIMA and 42% over a neural GRU. On the decision side the gain is structural. With unconstrained transfers every budget-exhausting policy ties on coverage and the robust formulation wins on routing; once transfers are scarce the tie breaks and the robust LP is never beaten on coverage. The main caveats are detour-scaled travel times, well-mixed populations, scaled cost units, and a single realised origin; more deeply, the per-region forecaster and co-monotone scenarios over-state the joint tail and understate transfers, which pay off most under *asynchronous* regional peaks. Correlated demand scenarios and a trust-level instance where exact solution warrants metaheuristics are the natural next steps.

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